UNIT 3 · 7 hours

Frequency Response and Frequency Domain Specifications



Frequency Domain Analysis

Control Systems (22EEG53)
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Learning objectives

- Define the frequency response of a system
- Define Mr, ωr, bandwidth and cut-off rate
- Derive Mr and ωr for a second-order system
- Compute the specifications from ωn and ζ

Syllabus context

Frequency domain analysis: frequency domain specifications peak resonance, resonant frequency, bandwidth and cut-off rate; correlation between time and frequency responses for second-order systems; Bode plot, gain margin and phase margin.

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Frequency response

Definition: steady-state output of an LTI system for a sinusoidal input of varying frequency.

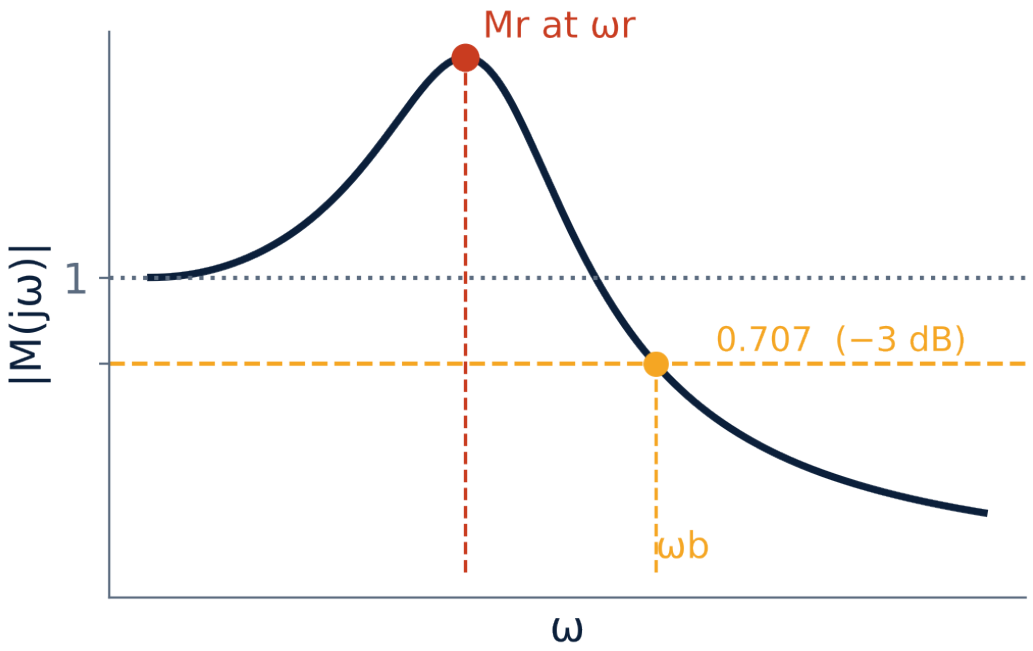
Obtain by putting $s = j\omega$ in the transfer function: $M(\omega)\angle\varphi(\omega) = G(j\omega)$.

Magnitude $M(\omega)$ = $|C/R(j\omega)|$ and phase $\varphi(\omega)$ are plotted against ω.

Advantages: easy to measure experimentally; stability and accuracy readable from plots; extends to systems whose transfer function is unknown.

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Frequency domain specifications



- Resonant peak Mr: maximum of $|M(j\omega)|$.
- Resonant frequency ωr: frequency at which Mr occurs.
- Bandwidth ωb: frequency where M falls to 0.707 (−3 dB) of its low-frequency value.
- Cut-off rate: slope of the magnitude curve beyond ωb (dB/decade).

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Second-order system

$$M(j\omega) = 1/\sqrt{[(1 - u^2)^2 + (2\zeta u)^2]}$$

u = ω/ωn

$$\omega r = \omega n \sqrt{(1 - 2\zeta^2)}$$

ζ ≤ 0.707

$$Mr = 1/(2\zeta\sqrt{(1 - \zeta^2)})$$

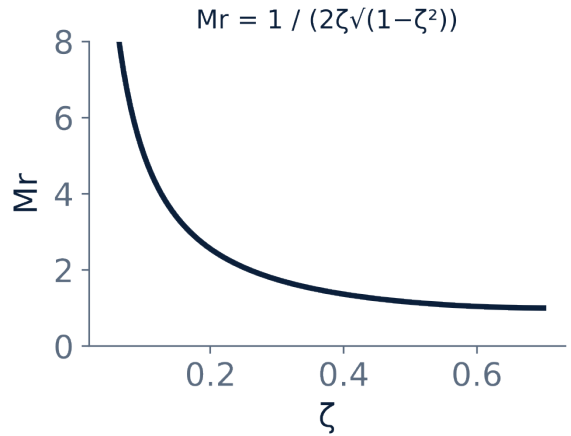
ζ ≤ 0.707

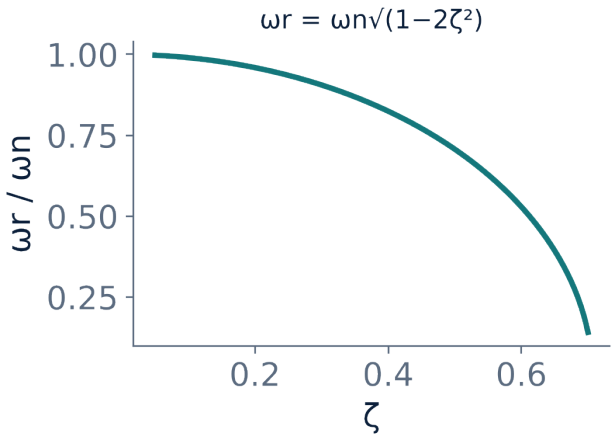
$$\omega b = \omega n[(1 - 2\zeta^2) + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}]^{\wedge}½$$

bandwidth

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Mr and ωr versus ζ





- Mr rises sharply as ζ → 0.
- Mr = 1 for ζ ≥ 0.707 (no peak).

- ωr falls from ωn to 0 as ζ → 0.707.

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Numerical: second-order specifications

Problem ωn = 10 rad/s, ζ = 0.4. Find ωr, Mr and bandwidth.

- $\omega_r = 10\sqrt{(1 - 2 \times 0.16)} = 10\sqrt{0.68} = 8.25$ rad/s
- $M_r = 1/(2 \times 0.4 \times \sqrt{(1 - 0.16)}) = 1/(0.8 \times 0.9165) = 1.364$
- $\omega_b = 10\,[0.68 + \sqrt{(0.1024 - 0.64 + 2)}]^{1/2} = 10\,[0.68 + 1.2093]^{1/2}$

Answer: ωr = 8.25 rad/s, Mr = 1.364 (2.7 dB), ωb = 13.75 rad/s

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Cut-off rate and bandwidth

- Cut-off rate: slope of the high-frequency asymptote: −20 dB/dec for 1st order, −40 dB/dec for 2nd order.
- Large bandwidth: fast rise time and good tracking, but also passes noise.
- Bandwidth ∝ ωn: ωb·tr ≈ constant (≈ 2.2 for first order).
- Specifications Mr and ωb are the frequency-domain equivalents of Mp and tr.

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Key takeaways

- Frequency response: $G(j\omega)$, magnitude and phase vs ω
- $\omega_r = \omega_n\sqrt{(1 - 2\zeta^2)}$, $M_r = 1/(2\zeta\sqrt{(1 - \zeta^2)})$
- Bandwidth: where M drops to 0.707 (−3 dB)
- Larger bandwidth ⇒ faster response

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Practice questions

Q1. Define Mr, ωr, bandwidth and cut-off rate.

Q2. Derive Mr and ωr for a second-order system.

Q3. ωn = 5, ζ = 0.3: find Mr and ωr.

Q4. How is bandwidth related to rise time?